

Zixuan (Austin) Huang

Open to Relocate | ah108@rice.edu | (346) 541-7952 | [LinkedIn](#)

EDUCATION

Rice University

September 2022 - May 2026

Bachelor of Science in Computer Science

GPA: 3.91/4.00

Relevant Coursework: Computational Thinking, Algorithmic Thinking, Program Design, Concurrent Program Design, Computer Organization, Reasoning about Algorithms

SKILLS

Languages: Python (3 yrs), C/C++ (3 yrs), Java (2yrs), HTML/CSS/Javascript, C#, SQL, MATLAB

Software & Operating Systems : Hadoop, AWS, Kafka, Spark, Microsoft Visio, TinyXML, Unity, Linux, Unix

Database: MySQL, Microsoft SQL, Vector Database

WORK EXPERIENCE

BlueVision LLC | **Software Engineer Intern** | Remote

June 2025 – Present

- Developed a C++ library to optimize camera aiming position and attitude for an image-mapping software solution
- Programmed with C++ Eigen library and implemented Newton's method to enhance the precision and efficiency of camera positioning, heading, and pitch calculations
- **Automated data workflows** for handling large datasets using **TinyXML**, for efficient reading, processing, and storage of data points, and constructed test packages with real-world data involving multiple edge cases
- Collaborated in software development process by utilizing Microsoft Visio to design sequence diagrams and refine class structure for the library

SeeU Education | **Software Engineer Intern** | Remote

March 2025 – May 2025

- Developed a WeChat Mini-Program (a web-based platform), enhancing accessibility to career consulting and graduate school application services, featuring functionalities such as event viewing, course purchases, and reservation bookings
- Designed the schema for the **MySQL** backend database for user features, such as user collections, user point systems and point logging system
- Programmed and dockerized backend API using **Java Spring Boot**, defined frontend **JavaScript** functions that call upon established backend APIs
- Tested the backend API call by deploying a local test database and constructing white and black box test cases

Occidental Petroleum | **IT Intern** | Houston, Texas

May 2024 – August 2024

- Developed an **ETL pipeline** to integrate inspection checklist data from digital twin platform Cognite Data Fusion into Microsoft **SQL** database GE APM
- **Automated data sync** every 4 hours with AWS CodePipeline and CodeBuild, boosting operational efficiency by 90%
- Standardized digital assets mapping between Cognite Data Fusion and GE APM for Occidental's digital twin initiatives
- Developed a process to extract coordinate data from EXR images, enabling 360-degree field view capability

Yupei Supply Chain Management | **Data and Technology Center Intern** | Shanghai, CN

May 2023 – July 2023

- Debugged and refactored 50+ old workflows for migration and integration within the new office automation system
- Supported system integration by generating flow diagrams and collaborating with developers to optimize performance
- Designed and implemented the user interface for the newly adopted **FLUX WMS storage system** using HTML

VOLUNTEER EXPERIENCE

J&J HiSoftware LLC | **Software Engineer** | Remote

Dec 2024 – Present

- Designed a **Python**-based database to store and manage uploaded job descriptions (JDs) and resume information, using **SQL** for data storage and retrieval
- Developed database structures to store parsed JDs and resume information for integration with upstream **data pipelines**
- Deployed local LLM to summarize resume information and improve accuracy of matching against **vector databases**
- Optimized database performance with teammates, supporting JDs and resume info data processing
- Embedded **vector database** in a local environment using **Python** to enable local testing and development without needing to access the remote database, migrated data from AWS to a newly created **Mavis database**

PROJECT EXPERIENCE

Unity video game development

April 2024 – November 2024

- Designed the enemy and health system for 2D top-down shooter "Runes" utilizing APIs and **creating custom classes**
- Evaluated new game features to verify expected effects and identified bugs within the game environment
- Utilized **GitHub** for code sharing and version control, creating forks to test features before merging into main repository